SAFETY DATA SHEET

Creation Date 29-Aug-2018 **Revision Date** 05-Sep-2019 **Revision Number** 2

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identification

Product Description: **Phenol:Chloroform:Isoamyl Alcohol (25:24:1)**
Cat No. : **J75831a**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company
Thermo Fisher (Kandel) GmbH .
Erlenbachweg 2
76870 Kandel
Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

E-mail address
tech@alfa.com
www.alfa.com
Product safety Tel + +049 (0) 7275 988687-0

1.4. Emergency telephone number

Carechem 24: +44 (0) 1235 239 670 (Multi-language emergency number)
Poison Information Center Mainz
www.giftinfo.uni-mainz.de Telephone: +49(0)6131/19240

Exclusively for customers in Austria:
Poison Information Center (VIZ)
Emergency call 0-24 clock: +43 1 406 43 43
Office hours: Monday to Friday, 8am to 4pm, tel: +43 1 406 68 98

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Acute oral toxicity	Category 3 (H301)
Acute dermal toxicity	Category 3 (H311)

Acute Inhalation Toxicity - Vapors	Category 3 (H331)
Skin Corrosion/Irritation	Category 1 B (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Germ Cell Mutagenicity	Category 2 (H341)
Carcinogenicity	Category 2 (H351)
Reproductive Toxicity	Category 2 (H361d)
Specific target organ toxicity - (repeated exposure)	Category 1 (H372)

Environmental hazards
Based on available data, the classification criteria are not met

2.2. Label elements

Signal Word **Danger**

Hazard Statements
H314 - Causes severe skin burns and eye damage
H341 - Suspected of causing genetic defects
H351 - Suspected of causing cancer
H361d - Suspected of damaging the unborn child
H372 - Causes damage to organs through prolonged or repeated exposure
H301 + H311 + H331 - Toxic if swallowed, in contact with skin or if inhaled
EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

Additional EU labelling
For use in industrial installations only

2.3. Other hazards

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS-No	EC-No.	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Phenol	108-95-2	EEC No. 203-632-7	50.0	Acute Tox. 3 (H301)

				Acute Tox. 3 (H311) Acute Tox. 3 (H331) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Muta. 2 (H341) STOT RE 2 (H373)
Chloroform	67-66-3	200-663-8	48.0	Acute Tox. 4 (H302) Acute Tox 3 (H331) Eye Irrit. 2 (H319) Skin Irrit. 2 (H315) Carc. 2 (H351) Repr. 2 (H361d) STOT RE 1 (H372)
Isoamyl alcohol	123-51-3	EEC No. 204-633-5	2.0	Flam Liq. 3 (H226) Acute Tox. 4 (H332) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335) (EUH066)

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Move to fresh air. Immediate medical attention is required.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
CO_2, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2), Hydrogen chloride.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe vapors or spray mist. Do not ingest.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and at the end of workday.

7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive 2006/15/EC of 7 February 2006 establishing a second list of indicative occupational exposure limit values in implementation of Council Directive 98/24/EC and amending Directives 91/322/EEC and 2000/39/EC on the protection of the health and safety of workers from the risks related to chemical agents at work. **UK** - EH40/2005 Containing the workplace exposure limits (WELs) for use with the Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended). Updated by September 2006 official press release and October 2007 Supplement. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority.

Component	European Union	The United Kingdom	France	Belgium	Spain
Phenol	TWA: 2 ppm (8h) TWA: 8 mg/m³ (8h) STEL: 4 ppm (15min) STEL: 16 mg/m³ (15min) Skin	STEL: 4 ppm 15 min STEL: 16 mg/m³ 15 min TWA: 2 ppm 8 hr TWA: 7.8 mg/m³ 8 hr Skin	TWA / VME: 2 ppm (8 heures). restrictive limit TWA / VME: 7.8 mg/m³ (8 heures). restrictive limit STEL / VLCT: 4 ppm. restrictive limit STEL / VLCT: 15.6 mg/m³. restrictive limit Peau	TWA: 2 ppm 8 uren TWA: 8 mg/m³ 8 uren STEL: 4 ppm 15 minuten STEL: 16 mg/m³ 15 minuten Huid	STEL / VLA-EC: 4 ppm (15 minutos). STEL / VLA-EC: 16 mg/m³ (15 minutos). TWA / VLA-ED: 2 ppm (8 horas) TWA / VLA-ED: 8 mg/m³ (8 horas) Piel
Chloroform	TWA: 2 ppm (8h) TWA: 10 mg/m³ (8h) Skin	TWA: 2 ppm TWA: 9.9 mg/m³ STEL: 6 ppm STEL: 29.7 mg/m³	TWA / VME: 2 ppm (8 heures). restrictive limit TWA / VME: 10 mg/m³ (8 heures). restrictive limit STEL / VLCT: 50 ppm. STEL / VLCT: 250 mg/m³. Peau	TWA: 2 ppm 8 uren TWA: 10 mg/m³ 8 uren Huid	TWA / VLA-ED: 2 ppm (8 horas) TWA / VLA-ED: 10 mg/m³ (8 horas) Piel
Isoamyl alcohol		STEL: 125 ppm 15 min STEL: 458 mg/m³ 15 min TWA: 100 ppm 8 hr TWA: 366 mg/m³ 8 hr	TWA / VME: 100 ppm (8 heures). TWA / VME: 360 mg/m³ (8 heures).	TWA: 100 ppm 8 uren TWA: 366 mg/m³ 8 uren STEL: 125 ppm 15 minuten STEL: 459 mg/m³ 15 minuten	STEL / VLA-EC: 125 ppm (15 minutos). STEL / VLA-EC: 458 mg/m³ (15 minutos). TWA / VLA-ED: 100 ppm (8 horas) TWA / VLA-ED: 366 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Phenol	TWA: 2 ppm 8 ore. Media Ponderata nel Tempo TWA: 8.0 mg/m³ 8 ore. Media Ponderata nel Tempo STEL: 4 ppm 15 minuti. Breve termine STEL: 16 mg/m³ 15 minuti. Breve termine Pelle	TWA: 2 ppm (8 Stunden). AGW - exposure factor 2 TWA: 8 mg/m³ (8 Stunden). AGW - exposure factor 2 Haut	STEL: 4 ppm 15 minutos STEL: 16 mg/m³ 15 minutos TWA: 2 ppm 8 horas TWA: 8 mg/m³ 8 horas Pele	huid TWA: 8 mg/m³ 8 uren	TWA: 2 ppm 8 tunteina TWA: 8 mg/m³ 8 tunteina STEL: 4 ppm 15 minuutteina STEL: 16 mg/m³ 15 minuutteina Iho
Chloroform	TWA: 2 ppm 8 ore. Media Ponderata nel Tempo TWA: 10 mg/m³ 8 ore. Media Ponderata nel Tempo Pelle	0.5 ppm TWA MAK 2.5 mg/m³ TWA MAK	TWA: 2 ppm 8 horas TWA: 10 mg/m³ 8 horas Pele	STEL: 25 mg/m³ 15 minuten TWA: 5 mg/m³ 8 uren	TWA: 2 ppm 8 tunteina TWA: 10 mg/m³ 8 tunteina STEL: 4 ppm 15 minuutteina STEL: 20 mg/m³ 15 minuutteina Iho
Isoamyl alcohol		TWA: 20 ppm (8 Stunden). AGW -	STEL: 125 ppm 15 minutos		TWA: 100 ppm 8 tunteina

		exposure factor 2 TWA: 73 mg/m^3 (8 Stunden). AGW - exposure factor 2 TWA: 20 ppm (8 Stunden). MAK TWA: 73 mg/m^3 (8 Stunden). MAK Höhepunkt: 40 ppm Höhepunkt: 146 mg/m^3	TWA: 100 ppm 8 horas		TWA: 370 mg/m^3 8 tunteina STEL: 150 ppm 15 minuutteina STEL: 550 mg/m^3 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Phenol	Haut MAK-KZW: 4 ppm 15 Minuten MAK-KZW: 16 mg/m^3 15 Minuten MAK-TMW: 2 ppm 8 Stunden MAK-TMW: 8 mg/m^3 8 Stunden	TWA: 1 ppm 8 timer TWA: 4 mg/m^3 8 timer Hud	Haut/Peau STEL: 5 ppm 15 Minuten STEL: 19 mg/m^3 15 Minuten TWA: 5 ppm 8 Stunden TWA: 19 mg/m^3 8 Stunden	STEL: 16 mg/m^3 15 minutach TWA: 7.8 mg/m^3 8 godzinach	TWA: 1 ppm 8 timer TWA: 4 mg/m^3 8 timer STEL: 3 ppm 15 minutter. value from the regulation STEL: 12 mg/m^3 15 minutter. value from the regulation Hud
Chloroform	Haut MAK-TMW: 2 ppm 8 Stunden MAK-TMW: 10 mg/m^3 8 Stunden	TWA: 2 ppm 8 timer TWA: 10 mg/m^3 8 timer Hud	Haut/Peau STEL: 1 ppm 15 Minuten STEL: 5 mg/m^3 15 Minuten TWA: 0.5 ppm 8 Stunden TWA: 2.5 mg/m^3 8 Stunden	TWA: 8 mg/m^3 8 godzinach	TWA: 2 ppm 8 timer TWA: 10 mg/m^3 8 timer STEL: 4 ppm 15 minutter. value calculated STEL: 15 mg/m^3 15 minutter. value calculated Hud
Isoamyl alcohol	MAK-KZW: 200 ppm 15 Minuten MAK-KZW: 720 mg/m^3 15 Minuten MAK-TMW: 100 ppm 8 Stunden MAK-TMW: 360 mg/m^3 8 Stunden	TWA: 100 ppm 8 timer TWA: 360 mg/m^3 8 timer	STEL: 80 ppm 15 Minuten STEL: 292 mg/m^3 15 Minuten TWA: 20 ppm 8 Stunden TWA: 73 mg/m^3 8 Stunden	STEL: 400 mg/m^3 15 minutach TWA: 200 mg/m^3 8 godzinach TWA: 100 mg/m^3 8 godzinach	TWA: 50 ppm 8 timer TWA: 180 mg/m^3 8 timer STEL: 75 ppm 15 minutter. value calculated STEL: 225 mg/m^3 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Phenol	TWA: 2 ppm TWA: 8 mg/m^3 STEL : 4 ppm STEL : 16 mg/m^3 Skin notation	TWA-GVI: 2 ppm 8 satima. TWA-GVI: 8 mg/m^3 8 satima. STEL-KGVI: 4 ppm 15 minutama. STEL-KGVI: 16 mg/m^3 15 minutama.	TWA: 2 ppm 8 hr. TWA: 8 mg/m^3 8 hr. STEL: 4 ppm 15 min STEL: 16 mg/m^3 15 min Skin	Skin-potential for cutaneous absorption STEL: 16 mg/m^3 STEL: 4 ppm TWA: 8 mg/m^3 TWA: 2 ppm	TWA: 7.5 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 15 mg/m^3
Chloroform	TWA: 2 ppm TWA: 10.0 mg/m^3 Skin notation	kože TWA-GVI: 2 ppm 8 satima. TWA-GVI: 10 mg/m^3 8 satima.	TWA: 2 ppm 8 hr. TWA: 9.8 mg/m^3 8 hr. STEL: 6 ppm 15 min STEL: 29.4 mg/m^3 15 min Skin	Skin-potential for cutaneous absorption TWA: 2 ppm TWA: 10 mg/m^3	TWA: 10 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 20 mg/m^3
Isoamyl alcohol	TWA: 360.0 mg/m^3 STEL : 450.0 mg/m^3	TWA-GVI: 100 ppm 8 satima. TWA-GVI: 366 mg/m^3 8 satima. STEL-KGVI: 125 ppm 15 minutama. STEL-KGVI: 458 mg/m^3 15 minutama.	TWA: 100 ppm 8 hr. TWA: 360 mg/m^3 8 hr. STEL: 450 mg/m^3 15 min STEL: 125 ppm 15 min		

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Phenol	Nahk TWA: 2 ppm 8 tundides. TWA: 8 mg/m^3 8 tundides. STEL: 16 mg/m^3 15	Skin notation TWA: 2 ppm 8 hr TWA: 8 mg/m^3 8 hr STEL: 16 mg/m^3 15 min STEL: 4 ppm 15 min	skin - potential for cutaneous absorption STEL: 4 ppm STEL: 16 mg/m^3 TWA: 2 ppm	STEL: 16 mg/m^3 15 percekben. CK TWA: 8 mg/m^3 8 órában. AK lehetséges borön	TWA: 1 ppm 8 klukkustundum. TWA: 4 mg/m^3 8 klukkustundum. Skin notation

	minutites. STEL: 4 ppm 15 minutites.		TWA: 8 mg/m^3	keresztüli felszívódás	Ceiling: 2 ppm Ceiling: 8 mg/m^3
Chloroform	Nahk TWA: 2 ppm 8 tundides. TWA: 10 mg/m^3 8 tundides.	Skin notation TWA: 2 ppm 8 hr TWA: 10 mg/m^3 8 hr	TWA: 10 ppm TWA: 50 mg/m^3	TWA: 10 mg/m^3 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 2 ppm 8 klukkustundum. TWA: 10 mg/m^3 8 klukkustundum. Skin notation Ceiling: 4 ppm Ceiling: 20 mg/m^3
Isoamyl alcohol	TWA: 100 ppm 8 tundides. TWA: 360 mg/m^3 8 tundides.		STEL: 125 ppm STEL: 450 mg/m^3 TWA: 100 ppm TWA: 360 mg/m^3	STEL: 1440 mg/m^3 15 percekben. CK TWA: 360 mg/m^3 8 órában. AK	TWA: 100 ppm 8 klukkustundum. TWA: 360 mg/m^3 8 klukkustundum. Ceiling: 200 ppm Ceiling: 720 mg/m^3

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Phenol	skin - potential for cutaneous exposure STEL: 4 ppm STEL: 16 mg/m^3 TWA: 2 ppm TWA: 8 mg/m^3	TWA: 2 ppm IPRD TWA: 8 mg/m^3 IPRD Oda STEL: 4 ppm STEL: 16 mg/m^3	Possibility of significant uptake through the skin TWA: 2 ppm 8 Stunden TWA: 8 mg/m^3 8 Stunden STEL: 16 mg/m^3 15 Minuten STEL: 4 ppm 15 Minuten	possibility of significant uptake through the skin TWA: 2 ppm TWA: 8 mg/m^3 STEL: 16 mg/m^3 15 minuti STEL: 4 ppm 15 minuti	Skin notation TWA: 2 ppm 8 ore TWA: 8 mg/m^3 8 ore STEL: 4 ppm 15 minute STEL: 16 mg/m^3 15 minute
Chloroform	skin - potential for cutaneous exposure TWA: 2 ppm TWA: 10 mg/m^3	TWA: 10 mg/m^3 IPRD TWA: 2 ppm IPRD Oda	Possibility of significant uptake through the skin TWA: 2 ppm 8 Stunden TWA: 10 mg/m^3 8 Stunden	possibility of significant uptake through the skin TWA: 2 ppm TWA: 10 mg/m^3	Skin notation TWA: 2 ppm 8 ore TWA: 10 mg/m^3 8 ore
Isoamyl alcohol	TWA: 5 mg/m^3	TWA: 5 mg/m^3 IPRD			TWA: 100 mg/m^3 8 ore STEL: 200 mg/m^3 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Phenol	TWA: 0.3 mg/m^3 0535 Skin notation STEL: 1 mg/m^3 0535	Ceiling: 16 mg/m^3 Potential for cutaneous absorption TWA: 2 ppm TWA: 8 mg/m^3	TWA: 2 ppm 8 urah TWA: 8 mg/m^3 8 urah Koža STEL: 4 ppm 15 minutah STEL: 16 mg/m^3 15 minutah	Binding STEL: 4 ppm 15 minuter Binding STEL: 16 mg/m^3 15 minuter TLV: 1 ppm 8 timmar. NGV TLV: 4 mg/m^3 8 timmar. NGV Hud	Deri TWA: 2 ppm 8 saat TWA: 8 mg/m^3 8 saat STEL: 4 ppm 15 dakika STEL: 16 mg/m^3 15 dakika
Chloroform	TWA: 5 mg/m^3 2098 Skin notation STEL: 10 mg/m^3 2098	Potential for cutaneous absorption TWA: 2 ppm TWA: 10 mg/m^3	TWA: 2 ppm 8 urah TWA: 10 mg/m^3 8 urah Koža	Indicative STEL: 5 ppm 15 minuter Indicative STEL: 25 mg/m^3 15 minuter TLV: 2 ppm 8 timmar. NGV TLV: 10 mg/m^3 8 timmar. NGV Hud	Deri TWA: 2 ppm 8 saat TWA: 10 mg/m^3 8 saat
Isoamyl alcohol	MAC: 5 mg/m^3		TWA: 370 mg/m^3 8 urah TWA: 100 ppm 8 urah STEL: 400 ppm 15 minutah STEL: 1480 mg/m^3 15 minutah		

Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Phenol			Total Phenol: 250 mg/g creatinine urine end of	: 120 mg/g Creatinine urine end of shift	Phenol (after hydrolysis): 120 mg/g

			shift		Creatinine urine (end of shift)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Phenol		Total phenol: 1.3 mmol/L urine after the shift.		Phenol: 200 µg/L urine at the end of exposure or end of work shift	total Phenol: 120 mg/g Creatinine urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Phenol			Phenol: 200 mg/L urine end of exposure or work shift		

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) No information available

Route of exposure	Acute effects (local)	Acute effects (systemic)	Chronic effects (local)	Chronic effects (systemic)
Oral				
Dermal				
Inhalation				

Predicted No Effect Concentration (PNEC) No information available.

8.2. Exposure controls

Engineering Measures
Ensure that eyewash stations and safety showers are close to the workstation location.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance		
Physical State	Liquid	
Odor	No information available	
Odor Threshold	No data available	
pH	No information available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	61 °C / 141.8 °F	
Flash Point	No information available	**Method -** No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available **Lower** 1.3 Vol % **Upper** 9.5 Vol %	
Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	Not applicable	Liquid
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Phenol	1.5	
Chloroform	2	
Isoamyl alcohol	1.28	
Autoignition Temperature	595 °C / 1103 °F	
Decomposition Temperature	No data available	
Viscosity	No data available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Heat.

10.5. Incompatible materials

Acids. Oxidizing agents.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2). Hydrogen chloride.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Product Information

(a) acute toxicity;
Oral Category 3
Dermal Category 3
Inhalation Category 3

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Phenol	LD50 = 340 mg/kg (Rat) LD50 = 317 mg/kg (Rat)	LD50 = 630 mg/kg (Rabbit)	LC50 = 316 mg/m³ (Rat) 4 h
Chloroform	LD50 = 450 mg/kg (Rat) LD50 = 695 mg/kg (Rat)	LD50 > 20 g/kg (Rabbit)	47,702 mg/L (Rat) 4 h
Isoamyl alcohol	LD50 = 1300 mg/kg (Rat)	LD50 = 3250 mg/kg (Rabbit) LD50 = 3970 µL/kg (Rabbit)	

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Phenol			Cat. 3B	
Chloroform				Group 2B

(g) reproductive toxicity; Category 2

(h) STOT-single exposure; No data available

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; Category 1

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Contains a substance which is:. The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Phenol	4-7 mg/L LC50 96 h 32 mg/L LC50 96 h	EC50: 10.2 - 15.5 mg/L, 48h (Daphnia magna) EC50: 4.24 - 10.7 mg/L, 48h Static (Daphnia magna)	EC50: 187 - 279 mg/L, 72h static (Desmodesmus subspicatus) EC50: 0.0188 - 0.1044 mg/L, 96h static (Pseudokirchneriella subcapitata) EC50: = 46.42 mg/L, 96h (Pseudokirchneriella subcapitata)
Chloroform	LC50: = 18 mg/L, 96h flow-through (Lepomis macrochirus) LC50: = 71 mg/L, 96h flow-through (Pimephales promelas) LC50: = 18 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: = 300 mg/L, 96h static (Poecilia reticulata)	EC50 = 28.9 mg/L/48h	EC50 = 560 mg/L/48h
Isoamyl alcohol	LC50 96 h 700 mg/L (rainbow trout)	EC50: = 260 mg/L, 48h (Daphnia magna)	EC50: = 181 mg/L, 96h (Desmodesmus subspicatus) EC50: = 493 mg/L, 72h (Desmodesmus subspicatus)

Component	Microtox	M-Factor
Phenol	EC50 21 - 36 mg/L 30 min EC50 = 23.28 mg/L 5 min EC50 = 25.61 mg/L 15 min EC50 = 28.8 mg/L 5 min EC50 = 31.6 mg/L 15 min	
Chloroform	Photobacterium phosphoreum: EC50 = 520 mg/L/5 min	

	Photobacterium phosphoreum: EC50 = 670 mg/L/15 min Photobacterium phosphoreum: EC50 = 670 mg/L/30min	
Isoamyl alcohol	EC50 = 2500 mg/L 17 h	

12.2. Persistence and degradability

Persistence Persistence is unlikely, based on information available.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Phenol	1.5	No data available
Chloroform	2	1.4 - 13
Isoamyl alcohol	1.28	No data available

12.4. Mobility in soil The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

12.5. Results of PBT and vPvB assessment No data available for assessment.

12.6. Other adverse effects

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalogue, Waste Codes are not product specific, but application specific.

Other Information Do not dispose of waste into sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN2922

14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s

Technical Shipping Name (PHENOL, CHLOROFORM)

14.3. Transport hazard class(es) 8

Subsidiary Hazard Class 6.1

14.4. Packing group II

ADR

14.1. UN number UN2922
14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s
Technical Shipping Name (PHENOL, CHLOROFORM)
14.3. Transport hazard class(es) 8
Subsidiary Hazard Class 6.1
14.4. Packing group II

IATA

14.1. UN number UN2922
14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s
Technical Shipping Name (PHENOL, CHLOROFORM)
14.3. Transport hazard class(es) 8
Subsidiary Hazard Class 6.1
14.4. Packing group II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required

14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
X = listed, Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), China (IECSC), Japan (ENCS), Australia (AICS), Korea (ECL).

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	IECSC	AICS	KECL
Phenol	203-632-7	-		X	X	-	X	X	X	X	KE-28209
Chloroform	200-663-8	-		X	X	-	X	X	X	X	KE-34076
Isoamyl alcohol	204-633-5	-		X	X	-	X	X	X	X	KE-23575

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Chloroform		Use restricted. See item 32. (see *http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT* for restriction details)	

National Regulations

Component	Germany - Water Classification (VwVwS)	Germany - TA-Luft Class
Phenol	WGK2	Class I : 20 mg/m³ (Massenkonzentration)
Chloroform	WGK3	Class I : 20 mg/m³ (Massenkonzentration)
Isoamyl alcohol	WGK1	

Component	France - INRS (Tables of occupational diseases)

Phenol	Tableaux des maladies professionnelles (TMP) - RG 14
Chloroform	Tableaux des maladies professionnelles (TMP) - RG 12
Isoamyl alcohol	Tableaux des maladies professionnelles (TMP) - RG 84

Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment.
Take note of Dir 94/33/EC on the protection of young people at work
Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H301 - Toxic if swallowed
H311 - Toxic in contact with skin
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H341 - Suspected of causing genetic defects
H351 - Suspected of causing cancer
H361d - Suspected of damaging the unborn child
H372 - Causes damage to organs through prolonged or repeated exposure
EUH066 - Repeated exposure may cause skin dryness or cracking
H226 - Flammable liquid and vapor
H302 - Harmful if swallowed
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H335 - May cause respiratory irritation

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
PNEC - Predicted No Effect Concentration
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC (volatile organic compound)

Key literature references and sources for data
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method

Environmental hazards	Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.

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End of Safety Data Sheet